HW: 4.3 Reproducible Data Analysis

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2026-04-15

Introduction

In this homework, I will demonstrate reproducible data analysis in R using Quarto. In the first example, I will use imports and cleans data on the busiest airports in the world. In the second example uses a Monte Carlo simulation to estimate the area under a curve.

Busiest Airports

In this section, I loaded and cleaned the data on the busiest airports from Wikipedia. The goal is to transform raw tables into a tidy data set that includes year, airport name, airport codes, and total passenger counts.

Table

Table 1: Passenger Traffic for Selected Airports (2020–2025)

year	airport	iata	icao	total_passengers
2020	Frankfurt Airport	FRA	EDDF	18768601
2021	Frankfurt Airport	FRA	EDDF	24812849
2022	Frankfurt Airport	FRA	EDDF	48918482
2023	Frankfurt Airport	FRA	EDDF	59355389
2024	Frankfurt Airport	FRA	EDDF	61564957
2025	Frankfurt Airport	FRA	EDDF	63189666
2021	Beijing Daxing International Airport	PKX	ZBAD	25051012
2024	Beijing Daxing International Airport	PKX	ZBAD	49441029
2025	Beijing Daxing International Airport	PKX	ZBAD	53618949
2020	Dubai International Airport	DXB	OMDB	25836771

year	airport	iata	icao	total_passengers
2021	Dubai International Airport	DXB	OMDB	29110609
2022	Dubai International Airport	DXB	OMDB	66069981
2023	Dubai International Airport	DXB	OMDB	86994365
2024	Dubai International Airport	DXB	OMDB	92331506
2025	Dubai International Airport	DXB	OMDB	95200000
2020	Tokyo Haneda Airport	HND	RJTT	31055210
2021	Tokyo Haneda Airport	HND	RJTT	25876429
2022	Tokyo Haneda Airport	HND	RJTT	50290705
2023	Tokyo Haneda Airport	HND	RJTT	78719302
2024	Tokyo Haneda Airport	HND	RJTT	85900167
2025	Tokyo Haneda Airport	HND	RJTT	91430000
2020	Dallas Fort Worth International Airport	DFW	KDFW	39364990
2021	Dallas Fort Worth International Airport	DFW	KDFW	62465756
2022	Dallas Fort Worth International Airport	DFW	KDFW	73362946
2023	Dallas Fort Worth International Airport	DFW	KDFW	81755538
2024	Dallas Fort Worth International Airport	DFW	KDFW	87817864
2025	Dallas Fort Worth International Airport	DFW	KDFW	85660127

Plot

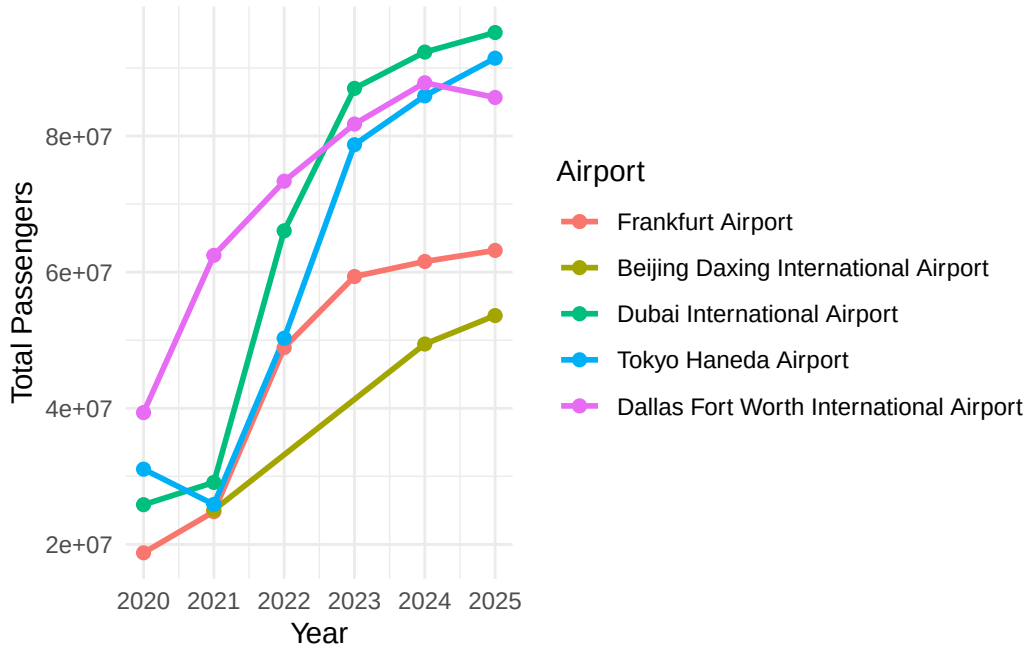


Figure 1: Passenger Traffic for Selected Airports

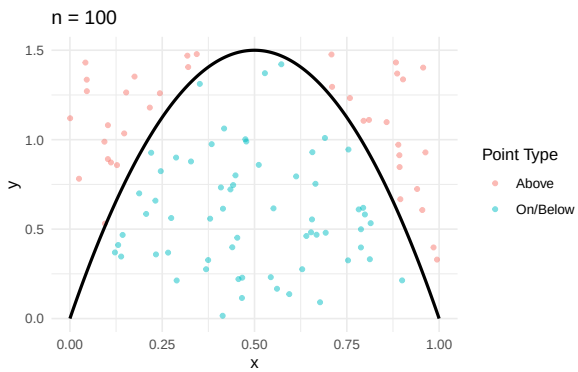
Interpretation

From the table and plot, it shows an increase in passenger traffic from 2020 to 2025 from the selected airports. Meaning, a pattern suggests a recovery in global air travel over time. The Dubai International Airport and Dallas Fort Worth International Airport consistently show high passenger totals, thus; supporting their importance as major international hubs. Beijing Daxing shows a noticeable growth in the later years, suggesting increasing demand and expansion. Overall, the visualization highlights both recovery and continued growth in airport activity. In conclusion, as an increase in passenger traffic increases so does recovery in global air travel.

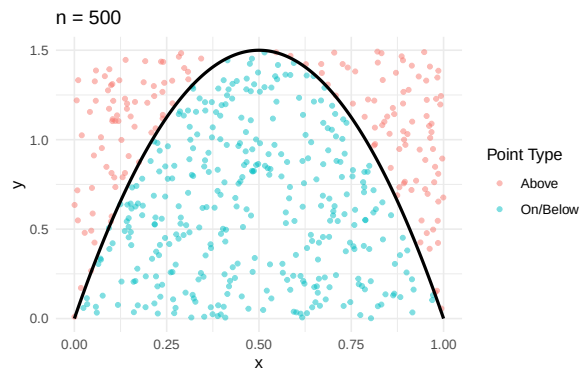
Monte Carlo Simulation

In this section, I used the Monte Carlo simulation in order to estimate the area under a curve. By selecting random points, I generated in a rectangle and the proportion of points that fall below the curve that is used to estimate the area.

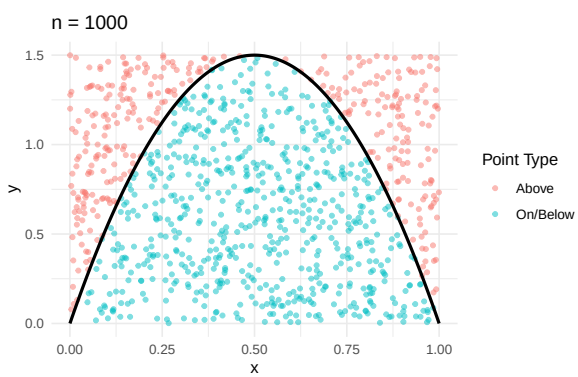
Plot



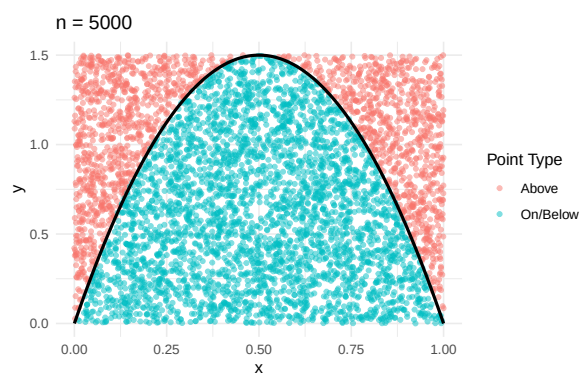
(a) Monte Carlo Simulation at Increasing Sample Sizes



(a) Monte Carlo Simulation at Increasing Sample Sizes



(a) Monte Carlo Simulation at Increasing Sample Sizes



(a) Monte Carlo Simulation at Increasing Sample Sizes

Interpretation

Monte Carlo simulation estimates an area by generating random points and checking whether they fall below the curve. In this example, the curve is based on a Beta(2,2) density, and the exact area under a probability density curve is 1. As the number of simulated points increases, the estimate becomes more stable and visually more convincing. Thus, the four panels show that smaller samples look more scattered, while larger samples produce a clearer approximation of the curve and support an estimate close to 1.

Conclusion

This assignment demonstrates how Quarto can combine code, figures, tables, and narrative writing into one reproducible report. Specifically, using the busiest airports example shows how real-world data can be cleaned and organized, while the Monte Carlo simulation illustrates how random sampling can be used to estimate mathematical quantities.

GenAI Usage Appendix

This appendix documents my use of generative AI tools in completing this assignment.

Usage 1

Tool used: ChatGPT
Version information: GPT-5.4 Thinking
Date of usage: April 15, 2026

Prompt(s) used: - Help me improve the organization and wording of my Quarto document. - Help me revise my narrative writing so it better explains the airport and Monte Carlo sections. - Help me fix rendering errors in my Quarto file.

Response: For improving the organization and wording of the Quarto document, the tool suggested structuring the report into clear sections (Introduction, Data Cleaning, Visualization, Interpretation, and Conclusion). It recommended writing concise topic sentences at the beginning of each section and ensuring that each paragraph clearly explains the purpose of the code and results. It also suggested improving transitions between sections so that the report reads as a continuous narrative rather than separate pieces.

For revising the narrative writing, the tool suggested clarifying the explanation of the busiest airports example by explicitly stating that the goal was to transform raw Wikipedia tables into a tidy dataset. It also recommended strengthening the interpretation by connecting observed trends (such as increases in passenger counts) to real-world meaning, like recovery in global air travel. For the Monte Carlo section, it suggested explaining the method step-by-step (random sampling, comparison to the curve, and estimation of area) and emphasizing that larger sample sizes produce more stable and accurate estimates.

For fixing rendering errors in the Quarto file, the tool explained that objects must be created before they are used in later chunks. Specifically, it identified that the error occurred because a dataset (such as airports_tidy) was referenced before it was defined. The solution was to move all data loading and cleaning code into an earlier setup chunk so that tables and plots could access the object during rendering. It also suggested ensuring all required libraries are loaded at the top of the document and that chunk execution order follows a logical sequence from data creation to output.

Code Appendix

```r
# ==========================================================
# Setup
# ==========================================================
rm(list = ls())

suppressPackageStartupMessages({
  library(tidyverse)
  library(rvest)
  library(janitor)
  library(readr)
  library(knitr)
})

# ==========================================================
# Busiest Airports Data
# ==========================================================
wiki_url <- "https://en.wikipedia.org/wiki/List_of_busiest_airports_by_passenger_traffic"

airport_tables <- wiki_url |>
  read_html() |>
  html_elements("table.wikitable") |>
  html_table(fill = TRUE)

years <- 2025:2020

yearly_airport_data <- map2(
  airport_tables[1:6],
  years,
  \(tbl, yr) {
    tbl |>
      clean_names() |>
      mutate(year = yr)
  }
) |>
  list_rbind()

airports_tidy <- yearly_airport_data |>
  select(year, airport, code_iata_icao, totalpassengers) |>
  mutate(
```

```r
    airport = str_squish(str_remove_all(airport, "\\[.*?\\]")),
    code_iata_icao = str_squish(str_remove_all(code_iata_icao, "\\[.*?\\]")),
    totalpassengers = parse_number(totalpassengers)
  ) |>
  separate_wider_delim(
    code_iata_icao,
    delim = "/",
    names = c("iata", "icao"),
    too_few = "align_start"
  )

selected_airports <- c(
  "Hartsfield-Jackson Atlanta International Airport",
  "Frankfurt Airport",
  "Beijing Daxing International Airport",
  "Dubai International Airport",
  "Tokyo Haneda Airport",
  "Dallas Fort Worth International Airport"
)

airports_tidy <- airports_tidy |>
  filter(airport %in% selected_airports) |>
  rename(total_passengers = totalpassengers) |>
  mutate(
    airport = factor(airport, levels = selected_airports)
  ) |>
  arrange(airport, year) |>
  select(year, airport, iata, icao, total_passengers)

# ==========================================================
# Airports Table
# ==========================================================
kable(airports_tidy)

# ==========================================================
# Airports Plot
# ==========================================================
ggplot(airports_tidy, aes(x = year, y = total_passengers, color = airport)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(
    x = "Year",
```

```r
    y = "Total Passengers",
    color = "Airport"
  ) +
  theme_minimal()

# ==========================================================
# Monte Carlo Helper Function
# ==========================================================
make_mc_plot <- function(n) {
  set.seed(123)

  x <- runif(n, min = 0, max = 1)
  y <- runif(n, min = 0, max = 1.5)

  curve_y <- dbeta(x, shape1 = 2, shape2 = 2)
  point_type <- ifelse(y <= curve_y, "On/Below", "Above")

  plot_data <- tibble(
    x = x,
    y = y,
    point_type = point_type
  )

  curve_x <- seq(0, 1, length.out = 200)
  curve_data <- tibble(
    x = curve_x,
    y = dbeta(curve_x, shape1 = 2, shape2 = 2)
  )

  ggplot() +
    geom_point(
      data = plot_data,
      aes(x = x, y = y, color = point_type),
      alpha = 0.5,
      size = 1.2
    ) +
    geom_line(
      data = curve_data,
      aes(x = x, y = y),
      linewidth = 1
    ) +
    coord_cartesian(xlim = c(0, 1), ylim = c(0, 1.5)) +
```

```r
    labs(
      title = paste("n =", n),
      x = "x",
      y = "y",
      color = "Point Type"
    ) +
    theme_minimal()
}

# ==========================================================
# Monte Carlo Plots
# ==========================================================
make_mc_plot(100)
make_mc_plot(500)
make_mc_plot(1000)
make_mc_plot(5000)
```